Example 4.3

Anbu bought 2 notebooks for ₹ 24. How much money will be needed to buy 9 such notebooks?

Solution Using unitary method we can solve this as follows:

$$\begin{aligned} \text{The cost of 2 notebooks} &= ₹\ 24 \\ \text{The cost of 1 notebook} &= \frac{24}{2} = ₹\ 12 \\ \text{Therefore, the cost of 9 notebooks} &= 9 \times ₹\ 12 \\ &= ₹\ 108 \end{aligned}$$

Hence, Anbu has to pay ₹108 for 9 notebooks.

Example 4.4

A car travels 90km in 2hours 30minutes. How much time is required to cover 210km?

Solution

$$\begin{aligned} \text{Time taken to cover 90 km} &= \text{2hrs 30mins} \\ &= 150 \text{ minutes} \\ \text{Time taken to cover 1 km} &= \frac{150}{90} \text{ minutes.} \\ \text{Time taken to cover 210 km} &= \frac{150}{90} \times 210 \text{ minutes} \\ &= 350 \text{ minutes} \\ &= 5 \text{ hours } 50 \text{ minutes} \end{aligned}$$

1 hour = 60 minutes
2 hours = 120 minutes

Thus, the time taken to travel 210 km is 5 hours 50 minutes

Exercise 4.1

1. Fill in the blanks

   (i) If the cost of 8 apples is ₹ 56 then the cost of 12 apples is _____.

   (ii) If the weight of one fruit box is $3\frac{1}{2}$ kg, then the weight of 6 such boxes is _____.

   (iii) A car travels 60 km with 3 liters of petrol. If the car has to cover the distance of 200 km, it requires _____ liters of petrol.

   (iv) If 7 m cloth costs ₹ 294, then the cost of 5 m of cloth is _____.

   (v) If a machine in a cool drinks factory fills 600 bottles in 5 hrs, then it will fill _____ bottles in 3 hours.

2. Say True or False

   (i) Distance travelled by a bus and time taken are in direct proportion.

   (ii) Expenditure of a family to number of members of the family are in direct proportion.

   (iii) Number of students in a hostel and consumption of food are not in direct proportion.

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